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Business Cycle Models with Endogenous Technology

By GEORGE W. STADLER *

This paper compares real and monetary business cycle models with and without endogenous technical change. If technology is endogenous, the properties of these models change significantly. In particular, both real and monetary models yield very similar output processes if growth is endogenous, and changes in aggregate demand can result in permanent changes in productivity, employment, and output. The effect of depreciation of technology is examined, and the pattern of real wage movements over the cycle when money wages are fixed but technology is changing is briefly considered. (JEL 131).

This paper examines the implications that endogenous technical change has for business cycle theory. Conventionally, both real and monetary models of the business cycle assume that the evolution of technology is exogenous to the economic system.¹ The monetary models ignore technical change, thereby implying that the demand-side disturbances that give rise to economic fluctuations in these models have no impact on technology. The real business cycle models assume that random changes in or “shocks” to productivity or technology cause output fluctuations in perfectly competitive environments. These temporary or permanent shocks to productivity are modeled as random drawings from a probability distribu-

tion: they are independent of economic forces and conditions and are thus exogenous to the economic environment.

However, there exists a substantial theoretical and empirical literature that emphasizes the dependence of technological progress on economic factors, implying that technology is, to a large degree, endogenous. The work of Kenneth Arrow (1962) points to the importance of learning by doing, and W. A. Eltis (1971) emphasizes the role of expenditure on research and development in generating economic growth. Jacob Schmookler’s (1966) pioneering study on the interrelationship between market conditions and innovation led him to conclude that “the belief that invention, or the production of technology generally, is in most instances a noneconomic activity, is false.”² James Utterback (1974) made a similar finding and reported:

Market forces appear to be the primary influence on innovation. From 60 to 80 percent of important innovations in a large number of fields have been in response to market demands or needs. The remainder have originated in response to new scientific or technological advances and opportunities.
[p. 621]

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¹Typical real business cycle models are those of John Long and Charles Plosser (1983) and Finn Kydland and Edward Prescott (1982). Robert King et al. (1988a) argue that the neoclassical model of capital accumulation (augmented by shocks to productivity) is the basic framework for real business cycle analysis, and discuss its limitations. The monetary models fall into two groups: the misperception models originating with the work of Robert Lucas (1973) and Robert Barro (1976), and models that emphasize nominal rigidities, such as Stanley Fischer (1977) or John Taylor (1980).

²Schmookler (1966, p. 208). Other studies have also found that economic factors have a significant influence on technology—Morton Kamien and Nancy Schwarz (1982) provide a survey of the literature.

Thus, exogenous changes in the scientific base (the “shocks” of real business cycle theory) do not appear to be the cause of most of the changes in productive techniques.

Conventional real and monetary business cycle theory (with exogenous technology) has ignored this body of literature. It is therefore interesting to see how real and monetary business cycle models with *endogenous* technology differ from the conventional models, and to what extent the results obtained from conventional models are sensitive to the assumption that technical change is independent of economic conditions.³

There is a second motivation for examining models of business cycles with endogenous technology. Charles Nelson and Charles Plosser (1982) first observed that most economic time-series appear to be nonstationary and are best modeled as containing stochastic trends. This implies that some fraction of innovations in output is permanent and alters the long-run trend of output. Traditionally, monetary innovations were considered to have only a transitory impact on output: the influence of a monetary shock dissipates over time, leaving the trend of output unchanged. Thus, models based on monetary innovations (or demand-side innovations generally) have difficulty in accounting for nonstationarity. This has fostered the growing belief that monetary

models of the business cycle are unable to provide a satisfactory explanation of output fluctuations and should be rejected in favor of real business cycle models.⁴

This paper shows that this inference is false. If technology is modeled as endogenous, a monetary business cycle model exhibits properties that are very different from those of the conventional models of Robert Lucas and Robert Barro or John Taylor and Stanley Fischer. There is a long-run non-neutrality of money in models with endogenous technology: monetary shocks have a permanent impact on output, and output is nonstationary, even in the absence of exogenous shocks to the supply side. The reason is that changes in demand that raise the level of output can, through a number of channels, exert a permanent influence on the supply side. For example, changes in the utilization of factor inputs when demand changes can result in reorganization and the acquisition of new skills; or a higher level of output may make innovation more profitable and result in the allocation of more resources to R&D. Furthermore, even real business cycle models with endogenous technology have somewhat different properties than conventional real business cycle models.

The following section develops a general model that contains both real and monetary disturbances. Unanticipated changes in the money supply have real effects in this model economy because nominal wages are as-

³A number of recent papers recognize the importance of endogenous technology in accounting for economic growth, for example, Lucas (1988), Edward Prescott and John Boyd (1987), and Paul Romer (1986), but these papers do not examine endogenous technology in a cyclical context. However, Robert King and Sergio Rebelo (1986) examine the impact of productivity shocks in a real business cycle model with endogenous growth, and their key conclusions are similar to those implied by the model of Section II, Part C below. This paper differs from King and Rebelo in that, first, they model the growth process as resulting from investment in human and physical capital, while I assume that changes in technological knowledge depend on the level of economic activity, specifically the level of labor input; second, this paper allows a role for money and aggregate demand shocks; third, it explicitly compares real and monetary business cycle models with and without endogenous growth. The model and conclusions of King and Rebelo's paper are briefly outlined in Section 3 of King et al. (1988b).

⁴For example, see Nelson and Plosser (1982, p. 159). Even Barro (1984) has suggested that the lasting contribution of the rational expectations revolution seems to lie in the development of the real, not monetary, business cycle theory. Bennett McCallum (1986), however, assesses the evidence against monetary models and concludes that it is insufficient to justify a rejection of this class of models. Furthermore, Kenneth West (1988) has shown that a purely monetary model with overlapping wage contracts can, for certain parameter values, yield an output process that has a root that is below but near unity. This implies that it may be impossible to discriminate in finite samples between a stationary monetary model and a nonstationary real business cycle model. By contrast, this paper demonstrates that with endogenous growth a purely monetary model can give rise to a nonstationary output process with a unit or greater-than-unit root.

sumed to be set in advance by one-period contracts. This is not a crucial assumption. Very similar results would be obtained if we assumed that prices, rather than wages, are sticky, or if a Lucas-type misperception framework had been adopted. Technology is partly endogenous because it is assumed that technical knowledge advances through learning by doing and depends on the level of labor input and on the level of labor productivity achieved in previous periods.

Within the general model are nested various sub-models that enable us to compare real and monetary models with and without endogenous technology. In Section II attention is focused on four sub-models; a pure monetary model with endogenous technology, a typical real business cycle model with exogenous technology, a real business cycle model with endogenous technology, and finally a hybrid model with transitory real and monetary shocks but no learning curve. The properties of the models are contrasted, and the long-run properties of the general model are examined.

The effects of depreciation of technology are examined in Section III. A monetary model with endogenous technology and a real business cycle model with exogenous technology are considered, where both the endogenous and exogenous technologies are subject to depreciation. The properties of a conventional real business cycle model change considerably if technology is subject to depreciation.

The fourth section considers the implications of learning for the pattern of real wages over the business cycle, when money wages are fixed by contract. Normally, a fixed money wage results in countercyclical movements in real wages, since prices move procyclically. However, once learning is introduced, the resulting pattern of real wages over the cycle becomes unclear. Even with nominal contracts, real wages may exhibit very little countercyclical movement.

The final section contains a summary and conclusions.

I. A General Model with Endogenous Technology

This section develops a simple model of output fluctuations with endogenous tech-

nology. Following Arrow, the concept of technical change I adopt comes from the basic premise that learning is the product of experience and takes place during activity.

I consider a simple closed economy. The supply side of the commodity market consists of a large number of competitive firms, all producing an identical good as output, that take the market price as given. Trading arrangements are sequential: at the beginning of each period, before the current period's price level is known, households enter into one-period nominal wage contracts with firms, agreeing to provide whatever amount of labor the firm requires at that wage rate. Thus, in order to negotiate the wage contract, agents must form expectations of both the price level and the representative firm's demand for labor before they know the value of the money supply (and aggregate demand) and before the real shocks hitting the economy are observed.

A. Aggregate Supply

The representative firm maximizes its stream of discounted expected profits:

$$V = \max_{\{L_t^i\}} E_t \left[\sum_{j=0}^{\infty} \beta^j \left(Y_{t+j}^i - \frac{W_{t+j}}{P_{t+j}} L_{t+j}^i \right) \right] \quad 0 < \beta < 1$$

where L_t^i is employment of labor by the representative firm, Y_t^i is the output of the representative firm, W_t is the money wage, P is the price level, and β is a discount factor. The production technology is assumed to be Cobb-Douglas:

$$(1) \quad Y_t^i = k (L_t^i)^\alpha Z_t^{(1-\alpha)} F_t \quad 0 < \alpha < 1$$

where k is a constant and F_t is a strictly positive stochastic shock that contains two components: (i) a permanent component, ξ_t , which can be thought of as a shock to technology, and (ii) a temporary productivity shock, η_t , so that in natural logarithms the evolution of F_t is given by

$$(2) \quad \begin{aligned} f_t &= \bar{f}_t + \eta_t \\ \bar{f}_t &= \bar{f}_{t-1} + \xi_t, \end{aligned}$$

where lowercase letters denote natural logarithms. Both η_t and ξ_t are stationary stochastic processes with zero mean, constant variance, and zero covariance.

Z_t is a scale factor that represents accumulated technical knowledge. It evolves according to

$$(3) \quad Z_t = Z_{t-1} \left[\frac{Y_{t-1}}{L_{t-1}} \right]^\lambda (L_{t-1})^\gamma$$

$$0 < \lambda, \gamma < 1.$$

The evolution of technical knowledge depends on the level of aggregate labor input (L_{t-1}) employed the previous period, and on the level of aggregate labor productivity attained the previous period.⁵ The greater the level of labor input, the greater is the scope for learning and acquisition of new skills. A higher level of labor input also requires more intensive use of factors fixed in the short run, thus raising the incentive to eliminate waste and bottlenecks. However, some change in productivity can occur independently of changes in labor input, for example, through reorganization of the production process and firm structure to achieve greater efficiency. The benefits of such changes are likely to persist for some time while the improved structures are in place, and to capture this source of productive improvements, I assume that technical knowledge also depends on the level of labor productivity attained in the previous period. For simplicity, it is assumed that Z is disembodied knowledge that does not depreciate (i.e., knowledge in books that is easily acquired by new entrants to the labor market). I abstract from physical capital (although Z is a kind of capital).

⁵An alternative but more complex way of introducing endogenous technology is to assume the firm faces an innovation production function and that innovation comes about through the application of resources to R&D. The results derived from such a model are similar to those presented below (see George Stadler, 1988). An alternative and simpler way of modeling learning is to assume that technical knowledge just depends on past levels of output, as in Stadler (1986).

Z depends on aggregate labor input and productivity. Since the individual firm is small, I assume it ignores the effect of its own employment decision and level of productivity on aggregate employment and productivity. The learning process can be thought of as a by-product of the output process. It is not necessarily under the conscious control of management. Furthermore, as labor moves between firms, learning skills become dispersed, so that aggregate labor input and productivity determine knowledge.

Assuming that the evolution of knowledge is exogenous to the firm enables us to treat the firm's problem like a one-period problem,⁶ and the firm's only decision variable is L_t^i . Taking the first-order condition for labor enables us to derive the representative firm's demand for labor (in logarithms):

$$(4) \quad l_t^{i,d} = \frac{\ln(\alpha k)}{1-\alpha} + \frac{1}{1-\alpha} (p_t - w_t) + z_t + \frac{1}{1-\alpha} f_t.$$

The labor supply function is assumed to be

$$(5) \quad L_t^s = e^{\phi_1} (W_t / P_t)^{\phi_2} \quad 0 < \phi_2 < 1.$$

At the beginning of the period the money wage is set at its expected market clearing level, that is, the wage at which the expected demand for labor during the period ahead equals the amount households expect to supply. The amount of labor households expect to supply will depend on the ex-

⁶Concavity of the production function (both labor and knowledge have diminishing marginal products) and the treatment of learning as an externality ensure that a competitive equilibrium exists each period with firms treating the state of technical knowledge as exogenously given. If learning were internalized, a competitive equilibrium might not exist. Since there is implicitly no limit to learning, utilization of labor yields increasing returns *over time* and the growth rate of output will rise over time. The assumptions that underpin this result are examined in Section II, Part E.

pected real wage. Taking the expectation of equation (5) yields, in logarithms,

$$(6) \quad E_{t-1}l_t^s = \phi_1 + \phi_2(w_t - p_t^e),$$

where $p_t^e \equiv E_{t-1}p_t$ is the rational expectation of the price level. The rational expectation of the firm's demand for labor formed at the beginning of the period is obtained by taking the expectation of equation (4), and using equations (2) and (3) to obtain

$$(7) \quad E_{t-1}l_t^{i,d} = \frac{\ln(\alpha k)}{1-\alpha} + z_t + \frac{1}{1-\alpha}(p_t^e - w_t) + \frac{1}{1-\alpha}\bar{f}_{t-1},$$

since $E_{t-1}z_t = z_t$ and $E_{t-1}f_t = \bar{f}_{t-1}$, because Y_{t-1} , L_{t-1} , Z_{t-1} , and $\bar{f}_{t-1}$ are contained in the information set available at the beginning of period t , and $E_{t-1}\eta_t = E_{t-1}\xi_t = 0$. Equating the expected supply and expected demand for labor given by the above two equations yields the money wage written into the labor contract:

$$(8) \quad w_t = p_t^e + \frac{\ln(\alpha k) - (1-\alpha)\phi_1}{1 + (1-\alpha)\phi_2} + \frac{(1-\alpha)z_t}{1 + (1-\alpha)\phi_2} + \frac{\bar{f}_{t-1}}{1 + (1-\alpha)\phi_2}.$$

Substituting equation (8) into equation (4) for w_t gives the firm's employment of labor:

$$(9) \quad l_t^i = a_0 + a_1(p_t - p_t^e) + a_2z_t + a_1f_t - a_3\bar{f}_{t-1}$$

$$a_0 = \frac{\ln(\alpha k)\phi_2 + \phi_1}{1 + (1-\alpha)\phi_2}$$

$$a_1 = \frac{1}{1-\alpha}$$

$$a_2 = 1 - \frac{1}{1 + (1-\alpha)\phi_2}$$

$$a_3 = \frac{1}{(1-\alpha) + (1-\alpha)^2\phi_2}.$$

Substituting (9) into a logarithmic form of the production function given by equation (1) and aggregating gives us the aggregate supply function:

$$(10) \quad y_t^s = q + b_0 + b_1(p_t - p_t^e) + b_2z_t + b_3(\xi_t + \eta_t) + b_4\bar{f}_{t-1}$$

$$b_0 = \ln(k) + \alpha a_0$$

$$b_1 = \alpha / (1 - \alpha)$$

$$b_2 = (1 - \alpha) + \alpha a_2$$

$$b_3 = 1 / (1 - \alpha)$$

$$b_4 = \frac{(1 + \phi_2)}{1 + (1 - \alpha)\phi_2},$$

where q is the natural logarithm of the number of firms in the economy. Hence aggregate supply will increase in response to a positive price surprise, a positive real shock, or a rise in technical knowledge.

B. Aggregate Demand and Expectations

I assume that aggregate demand is given by the quantity theory equation with velocity constrained to unity:

$$(11) \quad Y_t^d = M_t / P_t,$$

where Y_t^d is real aggregate demand and M_t is the money stock.

The money supply is assumed to follow a random walk with positive trend. The trend is set by the central bank so as to avoid an ongoing deflation in a growing economy, given that the velocity of money is constant. The money supply rule is public knowledge and, in logarithms, is

$$(12) \quad m_t = m_{t-1} + \mu + \varepsilon_t,$$

where μ is a constant trend and ε_t is a zero-mean stochastic error with constant variance.

When signing wage contracts, agents have at their disposal the past values of all rele-

vant variables (which enables them to calculate z_t), including $\bar{f}_{t-1}$. Writing (11) in natural logarithms, rearranging terms, and taking expectations gives us the rational expectation of the price level:

$$(13) \quad p_t^e = m_t^e - y_t^{de},$$

where $y_t^{de} \equiv E_{t-1} y_t^d$ is the rational expectation of aggregate demand formed at the beginning of the period. Equations (11) and (13) imply

$$(14) \quad (p_t - p_t^e) = (m_t - m_t^e) - (y_t^d - y_t^{de}).$$

Expected market clearing implies that $y_t^{de} = y_t^{se}$, so that one can take the expectation of equation (10) to obtain y_t^{de} :

$$(15) \quad y_t^{de} = y_t^{se} = q + b_0 + b_2 z_t + b_4 \bar{f}_{t-1},$$

since $E_{t-1}(p_t - p_t^e) = 0$, $\bar{f}_{t-1}$ is contained in the information set available at the beginning of the period, and the expected values of the real shocks (η_t and ξ_t) are zero. Actual market clearing implies $y_t^d = y_t^s$, so that subtracting equation (15) from equation (10) gives an expression for $(y_t^d - y_t^{de})$, the difference between actual and expected aggregate demand:

$$y_t^d - y_t^{de} = b_1(p_t - p_t^e) + b_3(\eta_t + \xi_t).$$

Substituting this expression into equation (14) and noting that $m_t - m_t^e = \varepsilon_t$ yields an expression for the difference between the actual and expected price level:

$$(16) \quad (p_t - p_t^e) = \frac{1}{1 + b_1} [\varepsilon_t - b_3(\eta_t + \xi_t)].$$

Thus prices diverge from their expected values each period because of unanticipated real and monetary shocks.

C. The Output Process

The level of output each period is obtained by substituting equation (16), which gives the price surprise into equation (10),

the aggregate supply function:

$$(17) \quad y_t = q + b_0 + \alpha[\varepsilon_t - b_3(\eta_t + \xi_t)] \\ + b_2 z_t + b_3(\eta_t + \xi_t) + b_4 \bar{f}_{t-1}.$$

Output depends on technology, both on the level of technical knowledge, z_t , and on the exogenous shocks to technology. However, real shocks are partly offset through price level effects. A positive value of η_t or ξ_t raises output and requires the price level to fall below its expected value to clear the commodity and money markets. The fall in the price level raises the real wage, thus counteracting the effect of the productivity shocks on employment and on output to some extent. Output also depends on unanticipated changes in the money supply, ε_t , and, as in all demand-side models of the business cycle with nominal wage contracting, an unanticipated rise in the money supply causes the price level to rise above its expected value and hence depresses the real wage and raises output.

The output process of equation (17) is nonstationary. The nonstationarity arises from two sources. First, as can be seen from equation (2), $\bar{f}_{t-1}$ is nonstationary because it consists of accumulations of permanent stochastic shocks. Second, z_t is nonstationary for, as can be seen by solving equation (3) backward, it depends on the cumulative sum of past levels of output and employment. The properties that each of these two sources of nonstationarity imposes on the output process become clearer in the next section, where a number of sub-models, containing some but not all the properties of the general model, are considered.

II. A Comparison of Real and Monetary Models with Endogenous and Exogenous Technology

Nested within the general model of equation (17) are at least four sub-models: a purely monetary model that contains no real shocks but with endogenous technology; real models with and without endogenous technology; and a hybrid model with monetary shocks and transitory real shocks but no learning. Each of these models is consid-

ered in turn, beginning with the monetary model with endogenous technology.

A. A Monetary Model with Endogenous Technology

This subsection considers a purely monetary model that contains no real shocks. The only disturbances to the economic system are unanticipated changes in the money supply, so that this model is a special case of the general model with $F_t = 1$ for all t (i.e., $\eta_t = \xi_t = 0$ for all t). The difference between actual and expected prices now only depends on innovations in the money supply:

$$(p_t - p_t^e) = \frac{1}{1 + b_1} \varepsilon_t,$$

and the output path is given by

$$(18) \quad y_t = q + b_0 + \alpha \varepsilon_t + b_2 z_t.$$

This output process is nonstationary and has a greater-than-unit root. This can be seen more clearly if one takes the first difference of output, uses equation (3) and an aggregate version of equation (9), and, after some manipulation, writes y_t as a moving average process:

$$(19) \quad y_t = (1 + \tau)^t y_0 + \alpha \varepsilon_t + \sum_{j=0}^{t-1} (1 + \tau)^j c_0 \\ + c_1 \sum_{j=0}^{t-2} (1 + \tau)^j \varepsilon_{t-j-1}$$

$$\tau = \left[1 - \frac{1}{1 + (1 - \alpha)\phi_2} \right] \gamma \\ + \left[\frac{1 - \alpha}{1 + (1 - \alpha)\phi_2} \right] \lambda, \quad \tau > 0;$$

$$c_0 = (b_2 - a_2)(\gamma - \lambda)q \\ + b_2(\gamma - \lambda)a_0 - a_2(\gamma - \lambda)b_0;$$

$$c_1 = [\tau\alpha + (1 - \alpha)(\gamma - \lambda)], \quad c_1 > 0,$$

where y_0 is a reference value in the past and I assume $\varepsilon_0 = 0$ for simplicity.

Equation (19) demonstrates that output depends on deterministic components of growth, captured by the constant, c_0 , and the term $(1 + \tau)^t y_0$. With endogenous technology the economy will grow even in the absence of stochastic shocks, because utilization of labor results in an ongoing learning process that increases technical knowledge, leading to higher productivity and output in future periods.

Output also depends on a nonstationary, stochastic growth component that consists of the cumulative sum of money-supply shocks that cause errors in price expectations. However, these errors have a permanent effect on output. Initially, a positive innovation in the money stock raises output by $\alpha \varepsilon_t$. In the absence of further shocks, output falls the following period, but not back to its previous level, for through learning a monetary innovation of ε_t raises output by $c_1 \varepsilon_t$ the next period. Thereafter, the impact of the shock gradually increases over time; in j periods ahead of it, output will have increased by $(1 + \tau)^{j-1} c_1 \varepsilon_t$. Thus, innovations in output are, to some degree, persistent, but the degree of persistence depends on the horizon from which the innovation is viewed: with endogenous technology, the impact of an innovation in output slowly rises over time.⁷ The reason for this is that any shock that causes an innovation in output adds a stimulus to an ongoing, endogenous growth process and raises the time profile of output permanently above

⁷Empirical studies have found differing estimates of persistence of innovations in output. John Campbell and Gregory Mankiw (1987) find that an unanticipated change in real GNP of 1 percent should change one's long-run forecast of output by *more* than 1 percent—between 1.3 and 1.9 percent. Other studies, such as John Cochrane (1988) and Mark Watson (1986), have found less persistence. Cochrane finds that at most 0.4 percent of an innovation in GNP is permanent, and Mark Watson finds 0.6 percent using an unobserved components model, but considerably more (1.7 percent) using an ARIMA model.

the level it would have achieved in the absence of the shock. A monetary shock at time t that raises employment and output above their steady-state values will, through learning, cause technical knowledge to rise above its steady-state value by time $t+1$. This rise in technology will increase the productivity of and demand for labor above their steady-state values at time $t+1$, bidding up the expected real wage and level of employment. Hence, as a result of a monetary innovation at time t , through learning, output will be above its steady-state value at time $t+1$ as well. However, the rise in labor input and output at time $t+1$ causes a further increase in technical knowledge above its steady-state value by time $t+2$. This results in a further rise in labor productivity and employment at time $t+2$, enabling further acquisition of technical skills by time $t+3$, which raises productivity and employment in that period, and so on. Thus, the endogenous growth process not only acts as a propagation mechanism for aggregate demand shocks but also amplifies their impact over time.

In contrast to conventional business cycle models, the sub-model captured by equation (19), like the general model it is derived from, has a greater-than-unit root in the output process (of value $(1+\tau)$) and is not difference stationary. Since (19) is in logarithmic form, the greater-than-unit root implies that in natural levels the growth rate of output will increase over time. This arises because, as discussed in Section II, Part E below, the growth rate of output essentially depends on the level of output.

Furthermore, the natural rate of unemployment is likely to depend on the history of aggregate demand in this model. The "natural" level of employment at any point in time can be regarded as the level of employment that prevails when there is no discrepancy between labor's actual and expected real wage. A sufficient condition for this is that there be no real or monetary surprises that drive a wedge between the actual and expected price level; that is, $\varepsilon_t = \xi_t = \eta_t = 0$. Aggregating equation (9) and using (3) to solve backward for z_t , we can

obtain the following expression for l_t :

$$(20) \quad l_t = q + a_0 + a_2 z_0 + a_2 \lambda \sum_{j=1}^t y_{t-j} \\ + a_2(\gamma - \lambda) \sum_{j=1}^t l_{t-j},$$

where z_0 is a reference value in the past. Thus, the natural level of employment at time t is a stochastic variable; it depends on the cumulative sum of past values of employment and on the cumulative sum of past values of output, which, from equation (19), depends on the cumulative sum of innovations in aggregate demand (and, in the more general model, on exogenous real shocks as well). The demand for labor and the equilibrium level of employment are consequently not invariant with respect to the history of aggregate demand in the economy. The natural rate of unemployment, being the difference between the supply of labor and the equilibrium level of employment, is likely to change in response to permanent changes in employment and the demand for labor.⁸ This suggests that in an economy with endogenous technology, changes in aggregate demand are likely to alter the natural rate of unemployment.

The results obtained from this model stand in strong contrast to the results derived from conventional monetary business cycle models. In both the misperception models and in the wage-contracting models, money has a transitory influence on eco-

⁸A general rise in the demand for labor resulting from a higher level of technical knowledge can reduce the natural rate of unemployment if it induces firms to establish training programs in areas of skill shortages. This could reduce the mismatch of skills required by employers and those possessed by the unemployed, hence lowering structural unemployment. A rise in demand for labor can also lead to greater expenditure by firms on recruitment and advertising, so leading to improved information flows and possibly lower search unemployment.

conomic activity: the impact of a monetary innovation dies away over time, and, in the long run, money is neutral. However, when technology is dependent on demand conditions, monetary shocks have a permanent impact on output. There is a long-run non-neutrality of money in models with endogenous technology.

B. *A Real Business Cycle Model with Exogenous Technology*

A second sub-model nested within the general model of equation (17) is a pure real business cycle model with technology evolving exogenously. Money has no impact on economic activity, and the model can be derived from the general model of equation (17) by assuming that not only the velocity but also the quantity of money is constant, so that the level of nominal aggregate demand is constant. In this case, unexpected price level changes are caused by technology shocks that alter real output. As is well known, the price level is countercyclical in such a model.

In the absence of endogenous technology, without a learning curve, $Z_t = 1$ for all t . Proceeding as before, the price surprise term reduces to

$$(21) \quad (p_t - p_t^e) = -[b_3 / (1 + b_1)](\eta_t + \xi_t),$$

for by assumption $\varepsilon_t = 0$ for all t , and the value of output is

$$(22) \quad y_t = q + b_0 - \alpha b_3(\eta_t + \xi_t) + b_3(\eta_t + \xi_t) + b_4 \bar{f}_{t-1}.$$

Output depends on exogenous real shocks. As in the general model, a positive innovation in η_t or ξ_t has a positive effect on output. The positive output effect is offset to some extent by the effect of the fall in the price level on the real wage. Taking the first difference of equation (22) and using (2) to

solve backward yields

$$(23) \quad y_t = y_0 + \eta_t + \xi_t + b_4 \sum_{j=1}^{t-1} \xi_{t-j},$$

where y_0 is a reference value at time $t = 0$ and I have assumed that $\eta_0 = \xi_0 = 0$.

Output is clearly nonstationary in this model, for it depends on the cumulative sum of permanent shocks to technology, the ξ 's. As in other real business cycle models, exogenous changes in technology cause permanent shifts in output, for the influence of the ξ 's does not decay over time. The initial impact of a positive innovation in ξ is, from equation (22), to raise output by $b_3(1 - \alpha)\xi_t$. In the following and all future periods, it raises output by $b_4\xi_t$, so that its long-run impact exceeds the short-run impact ($b_4 > (1 - \alpha)b_3$), for in the long run the offsetting rise in the real wage is absent.

C. *A Real Business Cycle Model with Endogenous Technology*

A third model of interest is a pure real business cycle model with endogenous technology, that is, a model in which learning occurs but money has no real effects. The production function of the representative firm is given by equation (1), and derivation of the model proceeds as in the case of the general model above, but with $\varepsilon_t = 0$ for all t , so that again the level of nominal aggregate demand is held constant. Thus, the only difference between this model and the preceding one is that technical knowledge, Z_t , rises over time in response to labor input and increases in productivity.

In the absence of monetary shocks, the difference between actual and expected prices depends only on real shocks and is given by equation (21). The output equation is very similar to that of the general model:

$$(24) \quad y_t = q + b_0 - \alpha b_3(\xi_t + \eta_t) + b_2 z_t + b_3(\xi_t + \eta_t) + b_4 \bar{f}_{t-1}.$$

Using equations (3) and (9) enables us to write, after some manipulation,

$$\begin{aligned}
 (25) \quad y_t = & (1 + \tau)^t y_0 \\
 & + \sum_{j=0}^{t-1} (1 + \tau)^j c_0 + \eta_t + \xi_t \\
 & + c_2 \sum_{j=0}^{t-2} (1 + \tau)^j \xi_{t-j-1} \\
 & + c_3 \sum_{j=0}^{t-2} (1 + \tau)^j \eta_{t-j-1} \\
 & + c_4 \sum_{j=0}^{t-1} (1 + \tau)^j \bar{f}_0 \\
 & + c_4 \sum_{i=0}^{t-2} \sum_{j=i}^{t-2} (1 + \tau)^j \xi_{t-j-1}
 \end{aligned}$$

$$c_2 = b_2 \lambda + b_4 + [b_2(a_3 - a_1) + a_2 b_4](\gamma - \lambda);$$

$$c_3 = b_2 \lambda; \quad c_4 = [b_2(a_1 - a_3) - a_2 b_4](\gamma - \lambda),$$

where $\bar{f}_0$ is a reference value in the past and I assume $\eta_0 = \xi_0 = 0$.

Output depends on the cumulative sums of the exogenous shocks and is nonstationary. It thus bears some resemblance to the conventional real business cycle model of equation (23). Nevertheless, there are two important differences between this model and the real business cycle model with exogenous technology. First, like the pure monetary model (and like the general model), this model has a greater-than-unit root, the value of the root being $(1 + \tau)$. The conventional model of equation (23) has a *unit* root, and is difference stationary, while the model with endogenous technology is not difference stationary. As in the monetary model, in a real business cycle model with endogenous technology the growth rate of output increases over time.

Second, with endogenous technology, transitory shocks to productivity (the η 's) have a permanent effect on output. Again, learning acts as a propagation mechanism for both demand-side and supply-side disturbances to output. A temporary rise in productivity stimulates learning and causes a rise in Z , shifting the long-run trend of output upward. Thus, in a model with endogenous growth, output will have a stochastic trend even if the only disturbances hitting the economy are transitory changes in productivity. (This is also the main finding of Robert King and Sergio Rebelo, 1986.)

D. A Hybrid Model

A final model that deserves consideration is one that contains both monetary shocks and transitory real shocks to technology but has no learning curve. Hence, $Z_t = 1$ for all t , and $F_t = e^{\eta_t}$, so that the firm's production function becomes

$$Y_t^i = k(L_t^i)^\alpha e^{\eta_t},$$

and the output process is governed by

$$(26) \quad y_t = q + b_0 + \alpha[\varepsilon_t - b_3 \eta_t] + b_3 \eta_t.$$

The trend, or expected future value of output conditional on current information, is

$$E_t y_{t+i} = q + b_0; \quad i > 1,$$

since $E_t \varepsilon_{t+i} = E_t \eta_{t+i} = 0$, for these are random variables with zero mean. It is at once apparent that this process is stationary, for output will only vary about the point $(q + b_0)$. This is true even if the monetary and real shocks have effects that persist for several periods before dying away, through, for example, the presence of adjustment costs. In the long run, money is neutral in this model. Thus, in a model with exogenous technology neither monetary shocks nor transitory shocks to technology have a permanent impact on output, thus highlighting the distinctions between models with endogenous and models with exogenous technology.

E. The Steady State

In the steady state neither real nor monetary shocks impinge on the economy— $\varepsilon_t = \eta_t = \xi_t = 0$ and $p_t = p_t^e$ for all t . The output process for the general model can then be written as a first-order difference equation:

$$(27) \quad y_t = (1 + \tau)y_{t-1} + c_0.$$

The solution to this difference equation is

$$y_{t+n} = [y_t + (c_0/\tau)](1 + \tau)^n - (c_0/\tau).$$

This equation describes an explosive growth process, for $\tau > 0$. The growth rate of the natural level of output, viewed n periods ahead at time $t + n$, will be $(1 + \tau)^n[y_t + (c_0/\tau)]\ln(1 + \tau)$, which increases as n rises. The rising growth rate occurs because the growth rate of technology, as expressed by equation (3), depends on the level of labor input and labor productivity attained. Hence the *growth rate* of technology essentially depends on its *level*, since the amount of labor utilized and labor productivity both depend on Z . This gives rise to the greater-than-unit roots that these models with learning contain; as the level of technology rises, so do productivity and labor input, causing the growth rate of technology to increase. The rise in Z results in a rising marginal product of labor that bids up the real wage and results in a rising level of employment. This can be seen from the long-run time paths of these variables:

$$z_t = (1 + \tau)z_{t-1} + c_z$$

$$\ln(\text{MPL}_t) = \ln(\alpha) + y_t - l_t$$

and

$$\Delta \ln(\text{MPL}_t) = (b_2 - a_2)[\tau z_{t-1} + c_z]$$

$$l_t - l_{t-1} = a_2(\tau z_{t-1} + c_z)$$

$$\Delta(w_t - p_t) = c_w(\tau z_{t-1} + c_z),$$

where $c_z = \lambda(q + b_0) + (\gamma - \lambda)a_0 > 0$; $c_w =$

$(1 - \alpha)/(1 + (1 - \alpha)\phi_2) > 0$; $(b_2 - a_2) > 0$. Thus the model exhibits rising growth rates of output, technical knowledge, productivity, employment, and real wages, since the growth rates of these variables depend on the growth rate of technology and thus on the level of technology.

These properties of the model may hold for considerable time periods and can account for the rising growth rates that Paul Romer (1986) reports for the past two centuries. However, these growth patterns may change substantially in the very long run, because they are based on the short-run labor supply function given by equation (5), and (5) is unlikely to hold in the very long run.

As society becomes wealthier, the income effect of a rise in wages may increasingly come to dominate the substitution effect, resulting in little or no increase in work effort as wages rise. Indeed, the long-run supply of labor schedule is probably backward-bending and the long-run value of ϕ_2 , the wage elasticity of labor supply, probably negative.⁹ However, the value of the root of output depends on the labor supply elasticity ϕ_2 , and τ falls as ϕ_2 falls.¹⁰ As $\phi_2 \rightarrow -\lambda/\gamma$, $\tau \rightarrow 0$, and if ϕ_2 takes the value $-\lambda/\gamma$, $\tau = 0$ and the output process changes to a nonexplosive unit-root process. In this case the time paths of the variables

⁹Daniel Hamermesh and Albert Rees conclude their discussion of labor supply by noting that "The long-run supply curve of employee hours to the economy as a whole... will probably be backward-sloping at the average real wage levels of developed countries.... The backward slope arises from the tendency in developed countries for hours actually worked per year to be reduced as real income rises over time while the overall rate of labor-force participation for all groups in the population taken together remains roughly constant or rises slightly." [1984, p. 82]

¹⁰ $\tau = [1 - (1/(1 + (1 - \alpha)\phi_2))]\gamma + [(1 - \alpha)/(1 + (1 - \alpha)\phi_2)]\lambda$, and $\partial\tau/\partial\phi_2 > 0$ if $\gamma > (1 - \alpha)\lambda$. This condition must hold if a rise in labor input is not, *ceteris paribus*, to have a negative or zero effect on technical knowledge, and is assumed to hold throughout. This can be seen by substituting an aggregate form of equation (1) lagged one period into (3) to obtain $Z_t/Z_{t-1} = k^\lambda Z_{t-1}^{(1-\alpha)\lambda} L_{t-1}^{\gamma-(1-\alpha)\lambda} F_{t-1}^\lambda$. If $\gamma < (1 - \alpha)\lambda$, then $\partial(Z_t/Z_{t-1})/\partial L_{t-1} < 0$.

become

$$y_t = y_{t-1} + c_0$$

$$z_t = z_{t-1} + c_z$$

$$\Delta \ln(\text{MPL}_t) = (b_2 - a_2)c_z$$

$$l_t - l_{t-1} = a_2 c_z$$

$$\Delta(w_t - p_t) = c_w c_z$$

and $a_2 < 0$ but $c_z, (b_2 - a_2), c_w > 0$ if $\phi_2 = -\lambda/\gamma$. In this case rising productivity is sufficiently offset by a falling supply of labor to dampen explosive growth.

Thus, if the wage elasticity is sufficiently negative, the model displays long-run features one might expect to observe in a developed economy with roughly constant population: declining employment, manifesting itself as a decline in average hours worked per person employed, steadily rising real wages and productivity, and technological advancement. If the wage elasticity rises above $-\lambda/\gamma$, then $\tau > 0$ and rising growth rates would manifest themselves again, though in conjunction with a decline in average hours worked per person employed if ϕ_2 is still negative. Hence, the actual growth path will depend crucially on the long-run dynamics of labor supply.¹¹

A further factor that can dampen the accelerating growth is limits to learning. Continuous learning can only occur if there are exogenous changes to the environment in which agents work. In the absence of new stimuli to learning, growth would slowly peter out. In terms of equation (3), γ and λ would approach zero over time: if $\gamma = \lambda = 0$, Z is constant and $\tau = c_0 = c_z = 0$, so growth ceases and the economy enters the stationary state.

Two pieces of evidence lend some support to my formulation of the learning pro-

cess and the output process that results from it. First, Romer (1986) presents empirical evidence for several industrial countries that *growth rates* of output have a positive trend and have been rising over the past two centuries. Second, in their survey of technical progress, Charles Kennedy and A. P. Thirlwall (1972) observe that "since product types are continually changing it is probably safe to assume that in aggregate there is no limit to the learning process."¹²

III. Depreciation of Technology

This section examines the properties of business cycle models if technology depreciates. Generally, technical and scientific knowledge is not regarded as subject to depreciation, except under abnormal circumstances such as times of war. However, one can think of certain technologies that, through changes in the state of nature, become obsolete before replacements are found for them. For example, in 1987, *The Economist* reported that "Bacteria are evolving resistance to antibiotics and insects resistance to pesticides faster than new chemicals are being invented. Before long, at this rate, medicine will revert to the days before penicillin and agriculture to the days before DDT."¹³ To examine the impact of depreciation in a model with endogenous technology, I allow Z to depreciate over time and replace (3) by

$$(28) \quad Z_t = Z_{t-1}^{(1-\delta)} \left[\frac{Y_{t-1}}{L_{t-1}} \right]^\lambda (L_{t-1})^\gamma \quad 0 < \delta < 1$$

where δ is the rate of depreciation of technology.

¹¹Changes in population growth will also affect the growth path of output, and a falling labor supply caused by declining population could also dampen accelerating growth. In this case, ϕ_1 in equation (5) would fall over time, causing the constants a_0 , b_0 , c_0 , and c_z to become time-varying.

¹²Kennedy and Thirlwall (1972, p. 39).

¹³March 21, 1987, p. 93. This prognosis was perhaps too pessimistic. In a more recent issue (April 16, 1988) *The Economist* suggests that, at least in agriculture, genetic engineering may overcome some of these problems, implying that the decline in agricultural technology is not permanent.

This section considers two models. The purely monetary model of Section II provides an example of the effect of depreciation in a model with endogenous technology. Second, a real business cycle model with depreciating exogenous technology is appraised. It emerges that such a real business cycle model cannot account for nonstationarity if technology is subject to depreciation.

A. A Monetary Model with Depreciation

The path of output in a pure monetary model is given by equation (18). Using equation (28) instead of equation (3) as the process that determines technical knowledge, one can write equation (18) as an ARMA process. Taking the first difference of equation (18) and substituting for l_{t-1} , yields, after some manipulation,

$$(29) \quad y_t = d_0 + \alpha \varepsilon_t - d_1 \varepsilon_{t-1} + (1 + \tau - \delta) y_{t-1}$$

$$d_0 = b_2(\gamma - \lambda)(q + a_0) - [a_2(\gamma - \lambda) - \delta](q + b_0)$$

$$d_1 = \alpha[1 + a_2(\gamma - \lambda) - \delta] - b_2(\gamma - \lambda).$$

This process may be stationary. A sufficient condition for stationarity is that $(1 + \tau - \delta) < 1$, or that $\delta > \tau$. Obviously, the higher δ , the rate of depreciation of technology, the more likely it is that this condition will hold and output will be stationary. Thus, once depreciation is introduced, the value of the root of output may be greater than unity, less than unity, or not significantly different from unity, and even a model with endogenous technology may yield a stationary output process.

B. A Real Business Cycle Model with Depreciation

This subsection considers the effect of depreciation in a conventional business cycle model with exogenous technology. The model is that given by equation (22), but

with one modification, namely, that the "permanent" technology shocks, the ξ 's, dissipate slowly over time, so that equation (2) is replaced by

$$(30) \quad f_t = \bar{f}_t + \eta_t$$

$$\bar{f}_t = (1 - \delta)\bar{f}_{t-1} + \xi_t,$$

and δ is again the depreciation rate. Using (30) to solve backward for $\bar{f}_{t-1}$ and substituting the result into the output equation (22) yields a stationary moving-average process for output:

$$(31) \quad y_t = q + b_0 + \eta_t + \xi_t + b_4 \sum_{j=0}^{\infty} (1 - \delta)^j \xi_{t-j-1}.$$

Shocks to output have a declining impact over time. Because $(1 - \delta) < 1$, the output process represented by this equation is stationary and has a less-than-unit root, unlike the model given by equation (22) without depreciation, which has a unit root.

While it is admittedly difficult to make a plausible case for the depreciation of technology in general, one cannot rule out the fact that some depreciation may occur in particular fields of knowledge. If technology does depreciate (δ is significantly greater than zero), conventional real business cycle models yield stationary output processes and cannot account for the observed nonstationarity of output or for persistence of innovations in output; only models with endogenous technology might display nonstationarity under these circumstances.

IV. Real Wages and Endogenous Technology

Empirical studies have generally failed to find significant countercyclical movements in the real wage. The most recent major study (Michael Keane et al., 1988) finds that real wages are mildly procyclical. This "stylized fact" is consistent with real business cycle models, but it appears to reject models with nominal wage contracts since

real wages move countercyclically in such models.¹⁴

This conclusion—that wage-contracting models are rejected by studies of real wage movements—is less easily drawn if technology is endogenous. Even if the money wage is fixed, real wages may not exhibit pronounced countercyclical movement, particularly if learning occurs rapidly in some markets.

The general model presented above was not constructed to explain the cyclical pattern of real wages. Nonetheless, examination of the simple labor market model posulated in Section I can yield some insights. The only change necessary is to introduce overtime rates and to assume that at the beginning of the period overtime rates are also set in money terms. In Figure 1, at an expected price level P_0 , a rise in employment above L_0 would result in overtime payments and raise the real and money wage. Hence the labor supply function is upward sloping, the slope depending on the structure of overtime pay.

A monetary shock that results in an unexpected rise in the price level reduces the real wage and the value of overtime payments, effectively shifting the labor supply schedule down and moving the labor market “equilibrium” from A to B. Employment (measured in man hours) rises: money wages rise through overtime pay, but by less than prices, so that the real wage falls below (W_0/P_0) . In a real business cycle model, an exogenous rise in productivity would shift the demand for labor function to the right (to L_1^d) and increase both employment and real wages. Price and wage flexibility ensures that the labor supply function does not shift when prices change, and the new labor market equilibrium is at point C.

The movement from A to B captures the conventional pattern of real wages and employment in a monetary business cycle model (in the Lucas-Barro models the rise in the labor supply results from misperception of

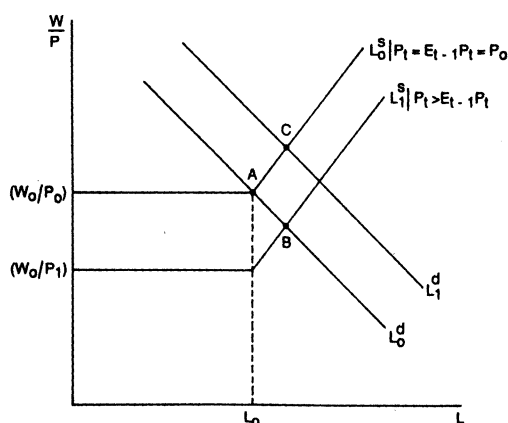


FIGURE 1. THE LABOR MARKET

the real return from working), but this pattern may not develop if technology depends on economic factors. In this case, as before, a monetary innovation boosts aggregate demand. Provided prices do not jump to a new higher level but adjust slowly (the excess money balances are spent gradually), then the labor supply function will begin to shift down slowly, raising employment. If the rise in employment results in a rapid increase in learning and innovation, the demand for labor will begin to rise. Thus, the downward movement of the labor supply schedule will set in motion an upward and rightward movement of the labor demand schedule. The resulting wage-employment pattern depends on the points of intersection of a downward-shifting labor supply function with an upward-shifting labor demand function. Rising prices reduce the real wage, but this is counteracted by rising productivity and output that place downward pressure on the price level and upward pressure on real wages; rising productivity retards the downward shift in the labor supply function. Clearly, a number of different wage paths are feasible, depending on the speed and magnitude of adjustment in the labor supply and demand functions in response to changing prices and productivity. It is conceivable that after a brief decline, real wages rise to their previous level. If there is a large increase in the demand for labor, one might even observe a *rise* in the real wage.

¹⁴This has led McCallum (1986) to suggest that price rather than wage stickiness must lie at the heart of cycles that are driven by aggregate demand shocks.

To conclude, real wage movements that are mildly procyclical but lagging *may* emerge from this model.¹⁵ Once learning is introduced, the pattern of real wages over the business cycle becomes unclear: it need not be strongly countercyclical, even if nominal wages are fixed by contract.

V. Summary and Conclusions

Empirical work suggests that many innovations in technology take place in response to market conditions, particularly demand conditions. It was shown that, in a simple model of endogenous technical change, disturbances to output that originate on the supply side or the demand side of the economy have markedly different effects than if technology is assumed to be independent of the economic process.

Endogenous technical change acts as a propagation mechanism for shocks that have only transitory effects in conventional models. A supply-side shock, such as a temporary rise in productivity, or a demand-side shock, such as an unanticipated rise in aggregate demand, can induce a permanent upward shift in the aggregate production function. Changes on the supply side of the economy are not independent of changes on the demand side, and there is a long-run non-neutrality of money in models with endogenous technology. The models derived in this paper also imply that the growth rate of output will increase over time provided the supply of labor does not fall sufficiently to choke off rising growth rates.

It was shown that if technology depreciates, conventional real business cycle models cannot account for the observed nonstationarity of output. However, both real and monetary models with endogenous technol-

ogy still yield nonstationary output processes provided the rate of depreciation of technical knowledge is not too high. Furthermore, the pattern of real wage movements is unclear if technical knowledge changes quite rapidly in response to changes in factor utilization. Even if money wages are fixed, real wages may display very little countercyclical movement.

There has been some debate as to whether models that emphasize aggregate demand innovations, or models that rely on real shocks as an impulse mechanism, provide a better characterization of output fluctuations. However, if technology is endogenous, real and monetary business cycle models exhibit very similar properties, as a comparison of equations (19) and (25) illustrates. Both classes of models can account for the persistence of innovations in output, and both can account for the rise in the growth rates of real output observed by Romer. This similarity makes it difficult, at the present time, to reject one model in favor of the other in terms of providing a better account of the output process.

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¹⁵The argument is made more formally in Stadler (1989). A question that is not satisfactorily addressed in the literature is the degree to which real wage changes are lagging over the business cycle. Victor Zarnowitz comments that "Money wages often rise less than prices in recoveries and more than prices in late expansion stages. There is a procyclical and lagging movement in labor costs per unit output." [1985, p. 529]

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